

Mohamed Ali Tarrouzi

Aspiring Computer Engineering Student Specialized in AI & Data Science

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PROFILE

5th-year engineering student at the Moroccan School of Engineering Sciences (EMSI), specializing in Artificial Intelligence and Data Science. Strong proficiency in AI, Machine Learning, Data Science, and Web Development. Seeking a Final Year Internship (PFE) starting in February 2027.

EDUCATION

State Engineering Degree in AI & Data Science — *2022 – Present*
EMSI (École Marocaine des Sciences de l'Ingénieur) — *Rabat, Morocco*

Baccalaureate in Physical and Chemical Sciences — *2021 – 2022*
Lycée M'hammed Guessous — *Rabat, Morocco*

PROFESSIONAL EXPERIENCE

DXC Technology Morocco — *July 2026 – August 2026*
Internship - Intelligent Secret Detection System
- Designed and developed an intelligent detection application using **FastAPI** and **React**.
- Configured Regex patterns and integrated **Claude** and **Groq** APIs via the **LiteLLM** gateway.
- Containerized with **Docker**, conducted testing, and managed integration and version control using **GitHub**.

Ministry of Justice — *July 2025 – August 2025*
Internship - Recommendation Management Platform
- Designed and developed a recommendation management platform using **Django**.
- Implemented dedicated roles for inspectors and agents to manage various recommendations across different entities.
- Developed the user interface using **HTML/CSS** and managed the **MSSQL** database.

ACADEMIC PROJECTS

Healthcare AI — Intelligent Medical Assistant (PFA)
- Developed a medical chatbot via **FastAPI**, **React**, and **PostgreSQL** leveraging **RAG** and **LLMs** (**Llama 3**, **Mistral**) for contextualized responses.
- Implemented a **Text-to-SQL** system enabling natural language queries on diabetic patient data to generate precise statistics.

Student Success Prediction
- Created a predictive model under **Python** based on **Machine Learning** algorithms (Decision Trees, Linear Regression) to analyze student success.
- Designed an interactive interface using **Streamlit** connected to a **MongoDB** database to ensure seamless data storage and visualization.

Sign Language Detection
- Developed a visual recognition system using **Python** and **OpenCV** to detect and interpret hand signs in real time.
- Trained and evaluated the model using a **Kaggle Dataset** to ensure precise and robust classification of various gestures.

School Management System
- Created a comprehensive web platform with **Django** and **Python** aimed at automating enrollments, grade management, and timetable optimization.
- Utilized the framework's business logic combined with an **SQLite** database backend to store institutional data.

TECHNICAL SKILLS

- **AI & Data Science:** Machine Learning, Deep Learning, Computer Vision, RAG Systems, LangChain, LangGraph.
- **Data Engineering & BI:** Apache Spark, Airflow, Airbyte, MinIO, PowerBI, Talend.
- **Databases:** ORACLE, MySQL, MSSQL, T-SQL, PostgreSQL, MongoDB (NoSQL).
- **DevOps & Tools:** Docker, Git, GitHub, CI/CD.
- **Back-End Development:** Python, Java, C#, .NET, FastAPI, Django.
- **Front-End Development:** TypeScript, JavaScript, React, HTML5, CSS3.

CERTIFICATIONS

- **Java SE 17 Developer** – Oracle Certification Program — *2026*
- **Introduction to Big Data** – UC San Diego (Coursera) — *2026*
- **Software Engineering: Design & Project Management** – Coursera — *2025*
- **Object-Oriented Programming in C++** – EPFL — *2025*

LANGUAGES & SOFT SKILLS

Arabic: Native | **French:** Fluent | **English:** Fluent | **German:** Basic knowledge
Soft Skills: Team player, Creative and solution-oriented, Rigorous and organized.